

Research Paper

Part decomposition and evaluation based on standard design guidelines for additive manufacturability and assemblability

Yosep Oh^{a,b}, Hyunwoong Ko^{a,c}, Timothy Sprock^d, William Z. Bernstein^d, Soonjo Kwon^{e,*}^a Associate, Engineering Laboratory, National Institute of Standards and Technology, Gaithersburg, MD 20899, United States^b Department of Physics, Georgetown University, Washington, DC 20057, United States^c School of Mechanical and Aerospace Engineering, Nanyang Technological University, 50 Nanyang Avenue, Singapore 639798, Singapore^d Engineering Laboratory, National Institute of Standards and Technology, Gaithersburg, MD 20899, United States^e Department of Mechanical System Engineering, Kumoh National Institute of Technology, Gumi-si, Gyeongsangbuk-do 39177, Republic of Korea

ARTICLE INFO

Keywords:

Part decomposition

Additive manufacturability

Assemblability

Evaluation

Optimization

Standard design guidelines

ABSTRACT

In comparison to traditional manufacturing, *Additive Manufacturing (AM)* provides unique capability to create complex part geometries. However, AM often suffers from its own limitations, such as the need for support structures, the presence of the stair-step effect and long build time. Recently, *Part Decomposition (PD)* techniques have been deployed to overcome these limitations. PD refers to the process of decomposing an original model into sub-assemblies. In the literature, PD studies have often proposed specific design considerations to evaluate that an original model is properly decomposed. For a comprehensive perspective, we address common and essential design considerations based on the standard design guidelines for AM in ISO/ASTM 52910. Since the standard design guidelines cover a variety of perspectives, we focus on the design considerations for *additive manufacturability* and *assemblability*. Then, evaluation indicators are defined for the design considerations as mathematical forms to quantitatively evaluate decomposed parts. The evaluation indicators are applied to the optimization of PD. As an optimization approach, a *genetic algorithm (GA)* is employed to recursively evaluate the outcomes of PD. In the GA, PD processes including *concave feature-based PD (CPD)* and *convex feature-based PD (VPD)* are considered to elaborately decompose an original model. In case studies, 9 models are compared with their decomposed parts for *material extrusion (ME)* and *vat photopolymerization (VP)* processes. On average, the build time decreases by 21% for ME and 70% for VP and the material consumption is reduced by 27% for ME and 22% for VP. Furthermore, when our optimization approach is compared to another similar method, *Near-convex Decomposition (NCD)*, the material consumption is reduced by an average of 26% and 62% for ME and VP processes, respectively.

1. Introduction

Additive Manufacturing (AM), commonly referred to as *3D printing*, is the process of joining materials, often layer-upon-layer, to produce parts from 3D model data, which differs from traditional manufacturing (*subtractive* and *formative*) [14]. The unique layer-upon-layer nature of AM has distinct benefits when compared with traditional manufacturing processes. AM facilitates the production of complex part geometries (e.g., freeform and lattice structures) by inherently providing design freedom to designers [43]. Moreover, AM offers newfound opportunities for developing functionally enhanced customized products [22,23]. However, AM suffers from its own limitations (e.g., the stair-step effect, presence of hollow spaces, long build times, and the need for support

structure).

Recently, *Part Decomposition (PD)* techniques have been deployed to overcome such limitations [36]. PD refers to the process of decomposing an original model (as-designed) into sub-assemblies. This is an opposite concept of *Part Consolidation (PC)* that is a re-design approach to consolidate multiple components into a single part [40,48]. While PC has been conventionally studied in *Design for AM (DfAM)* to minimize assembly time and cost, PD has been relatively recently emphasized [36]. In AM, PD is necessary for three primary reasons: (1) fitting in a limited build envelope [20], (2) minimizing build time and cost [35], and (3) improving quality [45]. To build a part that is larger than an AM machine, it should be decomposed. Moreover, by decomposing an original model and changing the build orientations of decomposed parts,

* Corresponding author.

E-mail addresses: soonjo.kwon@kumoh.ac.kr, soonjo.kwon.1@gmail.com (S. Kwon).<https://doi.org/10.1016/j.addma.2020.101702>

Received 24 July 2020; Received in revised form 5 November 2020; Accepted 6 November 2020

Available online 19 November 2020

2214-8604/© 2020 Elsevier B.V. All rights reserved.

we can avoid support structure and reduce build height that could cause more time and cost. Not only that but also we can reduce stair-stepping effect that is one of the main reasons causing poor surface quality in AM. While the three motives for AM have been highlighted in previous work [29,36,21], it is needed to underline evaluations for decomposed parts and to investigate their impact on PD decision-making. In this paper, the evaluation criteria for PD, named as *design considerations*, play a role in identifying whether an original model is properly decomposed.

The central aim of this study is to provide a comprehensive perspective by organizing design considerations for PD that can be applied to various types of AM processes. As such, our approach is based on standard design guidelines for AM in ISO/ASTM 52910 [15]. Since the standard guidelines in ISO/ASTM 52910 cover a broad range of aspects, we focus on design considerations for *additive manufacturability* and *assemblability*. The additive manufacturability is defined as the ability of materials to be joined layer-upon-layer together while minimizing manufacturing failures, defects, inefficiencies, or infeasibility due to geometries, materials, and processes [24]. The assemblability is defined as the ability of the part to assemble with a mating part and to perform a specified function [38]. While additive manufacturability focuses on a build process, assemblability aims at an assembly process that considers how decomposed parts are assembled. Based on the design considerations, we mathematically formulate *evaluation indicators (EIs)* to quantitatively assess the outcome of PD. The arrows in Fig. 1 represent relationships among the three categories: motives, design considerations, and EIs.

Furthermore, we apply the EIs to forming an objective function for an optimization routine. For the optimization of PD, a *genetic algorithm (GA)* is adopted. In the optimization process, two types of PD approaches are considered to decompose an original model: *concave feature-based PD (CPD)* and *convex feature-based PD (VPD)*. The optimization method is applied to our case studies. In the case studies, *material extrusion (ME)* and *vat photopolymerization (VP)* are considered. The main contributions of this study are summarized as follows:

- As witnessed in the comprehensive review of standard guidelines and PD studies, we identify which factors are critical in decomposing an object for AM.
- As mathematical reference models, the developed evaluation indicators can be applied to optimization and decision making for part decomposition.
- In terms of material consumption, we provide an advanced part decomposition method compared to *Near-convex Decomposition (NCD)* proposed by Demir et al. [10].

The rest of the paper is organized as follows. Section 2 gives literature reviews. Section 3 formulate EIs and Section 4 shows how the EIs

are applied to an optimization problem. In Section 5, a case study is demonstrated with discussion. Section 6 concludes this research and discusses future work.

2. Literature review

Here, we contextualize our work in relation to existing research on (i) design for AM approaches (Section 2.1), (ii) part decomposition methods (Section 2.2), and (iii) optimization for part decomposition (Section 2.3), and (iv) design considerations for PD instances for AM (Section 2.4).

2.1. Design for additive manufacturing

Design for Additive Manufacturing (DfAM) has become a core focus of researchers [2,44]. Design knowledge, tools, rules, processes, and methodologies for AM are substantially different from traditional *Design for Manufacturing and Assembly (DfMA)* principles [43]. An original design can be re-designed to realize various improvements uniquely enabled by AM such as reducing part weight and assembly time. In terms of the number of re-designed parts, the re-design methods can be categorized into three types [36]: (1) a consistent number of parts (e.g., topology optimization), (2) a decrease in the number of parts (i.e., part consolidation), and (3) an increase in the number of parts (i.e., part decomposition).

To help establish DfAM design guidelines, ISO/ASTM 52910 [15] classifies design considerations into eight categories: product, usage, sustainability, business, geometric, material property, process, and communication considerations. Among these classifications, the geometry considerations include surface roughness, minimum feature size, maximum aspect ratio, minimum feature spacing, recommended assembly fit, maximum part size, maximum unsupported feature, physical considerations, and mesh considerations.

In this paper, we particularly focus on the part decomposition among the re-design method categories. Then, we adopt the geometry considerations from ISO/ASTM 52910. We refer to those standard considerations as design considerations for additive manufacturability (shortened as manufacturability in the remaining sections) and assemblability.

2.2. Part decomposition methods

In the computer-aided design and computer graphics domains, part decomposition methods can be categorized into *cell-based decomposition* and *convex decomposition* [26]. The cell-based approach [39] decomposes a shape into simple cells and then recombines the cells to form several delta volumes. In this method, the original shape can be

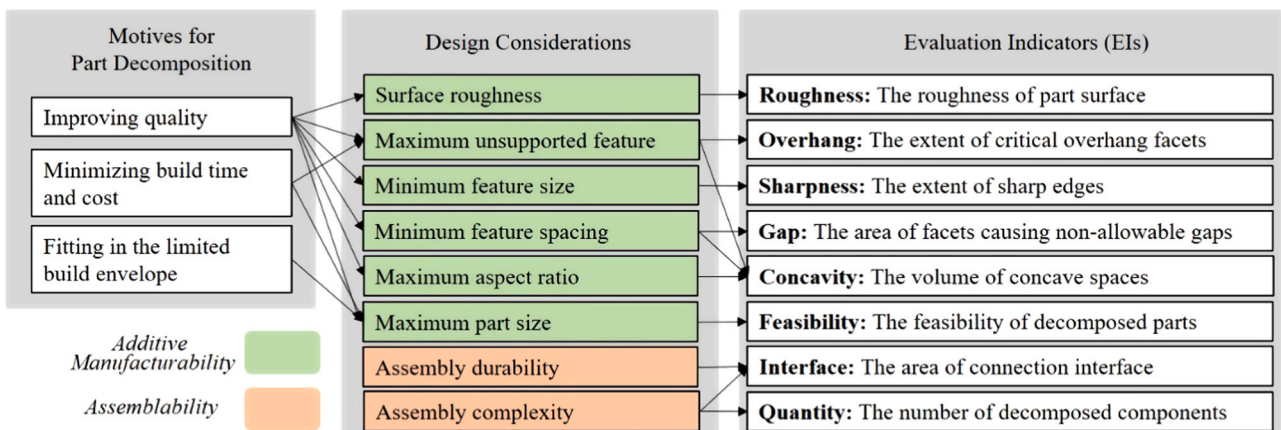


Fig. 1. Evaluation indicators (EIs) originated from the design considerations of ISO/ASTM 52910 [15].

represented by the Boolean union of the recombined volumes. However, the time required for recombining the cells exponentially increases with the number of cells. In addition, this method is regarded as more appropriate for dealing with polyhedrons rather than arbitrary curved shapes.

The convex decomposition method decomposes a shape into convex regions. *Exact Convex Decomposition (ECD)* [7] was proposed to decompose a shape into exact convex regions. However, ECD has been deemed impractical due to the unmanageable number of resulting components. To overcome this limitation, Lien and Amato [27,28] suggested *Approximate Convex Decomposition (ACD)* by relaxing the exact convexity constraint. Then, Mamou and Ghorbel [30] proposed a more efficient ACD method by dividing areas into clusters based on the concavity and other shape aspects. However, this approach has a critical downside in that the convex hulls of decomposed components may overlap with each other, which is unsuitable for destruction effects. In response, Mamou proposed *Volumetric Hierarchical ACD (V-HACD)* [31] that proceeds as follows. First of all, a voxelization or tetrahedralization process is applied in order to approximate the volume based on a set of primitives (i.e., voxels or tetrahedra). Then, the generated primitives are recursively decomposed according to a top-down hierarchical approach. This decomposition process is continuously applied to the created sub-parts until a concavity threshold is reached. To guarantee non-overlapping convex hulls, V-HACD decomposes an object according to a set of clipping planes based on concavity, balance, and symmetry criteria.

In this paper, we adopt V-HACD as one of the main PD methods since it is an efficient algorithm to compute convex approximations. However, V-HACD has two critical issues in terms of manufacturability. First, the outcomes of this algorithm are convex hulls that deviate from the geometry of the original model. This may not be acceptable in a manufacturing perspective pursuing part accuracy and precision. Second, once decomposed components become convex (or near-convex [10]) parts and have few concave features, the algorithm cannot decompose parts further. This is largely because the algorithm concentrates on removing concave features. This could be problematic when the outcome of this algorithm is larger than the build envelope of an AM machine. To deal with these downsides, we include a post-processing phase after the V-HACD algorithm runs.

2.3. Optimization for part decomposition in AM

The decomposition algorithms deployed in PD studies often use a greedy approach [20,35]. As a result, when a part is decomposed by a cutting plane, a local optimum is chosen from a number of cutting candidates at each decomposition stage. In many cases, this greedy strategy is computationally efficient but does not guarantee the global optimum [9]. To overcome this limitation, some studies [8,29] adopted a *beam search* that maintains a set of best candidates at each decomposition stage.

Another approach is to allow the PD algorithm to continue until certain considerations are met (e.g., running time of algorithm or number of repeats). Implementing the same concept, Demir et al. [10] repeatedly invoked V-HACD while changing input parameters at each iteration. To find the optimal input parameters, the authors adopted *Bound Optimization BY Quadratic Approximation (BOBYQA)*, a derivative-free optimization based on a trust-region method [37]. Inspired by the work of Demir et al. [10], we adopt a similar optimization method, iteratively calling V-HACD and changing its input parameters. However, instead of BOBYQA, a genetic algorithm is applied to avoid a situation where the solution is dominated too heavily by the default input parameters.

In this study, to optimize PD, we chose four parameters (alpha, beta, gamma, and maximum concavity) from the 12 input parameters of V-HACD because they mainly affect the number and geometry of decomposed parts. The selected four parameters are different from the three

parameters (resolution of voxelization, minimum volume per convex hull, and maximum number of vertices per convex hull) that Demir et al. [10] selected for V-HACD. This is largely because controlling the three parameters of Demir et al. could generate convex hulls that are smaller than the original geometries, which does not make logical sense.

2.4. Design considerations for part decomposition in AM

When applying PD methodologies, AM studies have addressed a variety of design considerations for specific motives and objectives. Luo et al. [29] proposed a PD method (*Chopper*) for AM and inspired other PD studies [8,35]. For PD, they considered different objective functions for various design considerations including the number of parts, connector feasibility, structural soundness, and aesthetics. Yao et al. [49] decomposed an original model based on three motives (printability, fast production, and space-saving of the envelope volume) and considered six attributes (stress load, surface details, interface area, packed size, printability, and assembling). Jiang et al. [20] considered the size, symmetry, and assemblability of decomposed parts. Demir et al. [10] included concavity, surface angles, a balance between size and number of components, and deviation from the original model. Oh et al. [36] surveyed and organized previous PD studies in terms of printability, productivity, functionality, artistry, and interoperability. While some studies spanned multiple objectives and attributes, others often concentrated on a specific challenge (e.g., reducing support structures). Wu et al. [47] presented a support-effective PD method by printing different model regions depending on various directions. Karasik et al. [21] proposed a support-free PD method based on object partitioning for a ME printer.

The common drawbacks of the previous studies are that they fail to focus on comprehensive perspectives of manufacturability and assemblability for AM. This is why we refer to standard design guidelines and recommendations for AM (i.e., ISO/ASTM 52910). Moreover, a lot of technical terms about design considerations are used inconsistently in the literature. To cope with this, the definition and meaning of terms are organized based on the standard guidelines in Section 3.

3. Design evaluation indicators for decomposed parts

In this section, we define and formulate the EIs of design considerations based on ISO/ASTM 52910 [15]. In most cases, the original meaning of the design considerations is narrowed down to formally characterized EIs. Table 1 presents eight EIs stemmed from the design considerations of ISO/ASTM 52910. When a build orientation or a user-defined threshold is required to formulate EIs, it is marked as O in Table 1. Otherwise, it is marked as X in the table. The following sub-sections detail the corresponding EI.

3.1. Roughness: the roughness of part surface

ISO/ASTM 52910 [15] defines surface roughness as a measure of the deviations normal to the surface of an additively manufactured surface from the ideal surface (i.e., surface texture). Also, according to ISO 4287 [13], the concept of surface texture broadly covers waviness and

Table 1
Design evaluation indicators (EIs).

EIs	Type	Formula	Orientation	Threshold
Roughness	Manufacturability	Eq. (1)	O	X
Overhang	Manufacturability	Eq. (2)	O	O
Sharpness	Manufacturability	Eq. (3)	X	O
Gap	Manufacturability	Eq. (4)	X	O
Concavity	Manufacturability	Eq. (5)	X	X
Feasibility	Manufacturability	Eq. (6)	O	O
Interface	Assemblability	Eq. (7)	X	X
Quantity	Assemblability	Eq. (8)	X	X

primary profile as well as roughness. In the AM literature, numerous studies [1,4,45] narrowed down the meaning of surface roughness into the upward- and downward-facing surface roughness for varying angles caused by discrete steps (i.e., stair-steps [16]). We accept this specified meaning to formulate surface roughness for AM.

A variety of factors including build orientation, feedstock, and process parameters (e.g., layer thickness) affect the surface roughness. However, for computational efficiency at the design evaluation stage, we adopt a way to roughly calculate the surface roughness by considering the part geometry information and the build orientation. Even though the surface roughness is formulated in various ways [1,4,45], a key idea is to multiply the surface area with the angle between the surface normal and the build direction. For PD evaluation, we formulate *Roughness*, G_1 , as the sum of the surface roughness of facets for all decomposed components, which is expressed as follows:

$$G_1(C) = \sum_{c \in C} \sum_{f \in c} E^{SR}(f) A(f) \quad (1)$$

$$E^{SR}(f) = \begin{cases} |\cos(\theta_f^{SR})| & \text{if } 0 < \theta_f^{SR} < \pi \\ 0 & \text{otherwise} \end{cases}$$

where, C is a set of decomposed components ($c \in C$); f is a facet of a component; $A(f)$ is the area of Facet f ; and θ_f^{SR} is the deviation angle between the face normal vector and the build orientation ($0 \leq \theta_f^{SR} \leq \pi$).

The surface roughness for each facet (i.e., a polygon) is measured by multiplying the normalized deviation degree, $E^{SR}(f)$, with the area of Facet f , $A(f)$. The absolute value and cosine for θ_f^{SR} are used for normalization by scaling between 0 and 1. The build orientation is defined as the opposite direction of the face normal of the base plane. Build orientations can be determined in various ways, such as placing the largest facet as the base plane [10] or finding the shortest build height [33].

Minimizing G_1 can be an objective function to improve surface quality. In this case, build orientation could be a decision variable. Fig. 2 represents how G_1 decreases through PD. In this figure, the stair-step effect is reduced by re-orienting the build direction of decomposed parts.

3.2. Overhang: the extent of critical overhang facets

Support structure is often generated in AM processes. In AM processes, support structure is mainly required for two reasons [19]: (1) to prevent thermal deformation and (2) to keep standing balance of parts. We focus on the second reason that is usually caused by overhang facets which are downward-facing surfaces [25].

Depending on the angle with respect to the vertical axis, some overhang facets are self-supported. In the literature [5,12,18], 45° is often set as the threshold to determine whether support structures are created. ISO/ASTM 52910 [15] refers to the concept of the maximum unsupported feature as the largest feature size that can be built at a certain threshold angle without support structures. This design

consideration is meant to minimize the amount of support structure.

Originated from the maximum unsupported feature, *Overhang*, G_2 , refers to the sum of the extent of critical overhang facets for all decomposed components. Herein, the critical overhang facets are downward faces exceeding the threshold angle. The amount of support structure is often estimated as the volume of support structure in a direct way [12,46]. However, to simplify the formulation, we indirectly estimate the amount of support structure as critical overhang facets as follows:

$$G_2(C) = \sum_{c \in C} \sum_{f \in c} E^{OV}(f) A(f) \quad (2)$$

$$E^{OV}(f) = \begin{cases} |\cos(\theta_f^{OV})| & \text{if } 0 \leq \theta_f^{OV} \leq \tau^{OV} \\ 0 & \text{otherwise} \end{cases}$$

where, τ^{OV} is the threshold to determine whether support structures are caused; and θ_f^{OV} is the deviation angle between the face normal vector of Facet f and the face normal of the base plane (see Fig. 3-(a)). The range of θ_f^{OV} is defined as $0 \leq \theta_f^{OV} < \frac{\pi}{2}$.

The extent of critical overhang Facet f is calculated by multiplying the normalized deviation degree, $E^{OV}(f)$, with the area of Facet f , $A(f)$. As used for θ_f^{SR} , the absolute value of the cosine for θ_f^{OV} is for normalization. Even though we define θ_f^{OV} as a constant parameter, it practically depends on feature size or length.

Minimizing G_2 can be an objective function to prevent support facets. Fig. 3-(b) shows how to reduce potential support structures throughout the PD process. In this figure, critical overhang facets are prevented by re-orienting the build direction of the decomposed parts.

3.3. Sharpness: the extent of sharp edges

ISO/ASTM 52910 [15] refers to the minimum feature size as the smallest feature that a build process is capable of producing. This design consideration is introduced to prevent tiny features from being produced in a build process. These tiny features can be of various types including both positive volumes (e.g., ribs and bosses) or negative volumes (e.g., holes). In this paper, we focus on sharp edges including positive and negative edges as shown in Fig. 4.

Similar to the concept of sharp edges, the sharpness of vertices can be formulated based on curvatures on meshes [32,42]. To simplify formulation, we suggest a method for calculating sharpness based on the edge between two contact facets. Here, we define *Sharpness*, G_3 , as the sum of the length of sharp edges for all decomposed components, which is expressed as follows:

$$G_3(C) = \sum_{c \in C} \sum_{e \in c} E^{SH}(e) \quad (3)$$

$$E^{SH}(e) = \begin{cases} 0 & \text{if } \tau^{SH} < \theta_e^{SH} \\ L(e) & \text{otherwise} \end{cases}$$

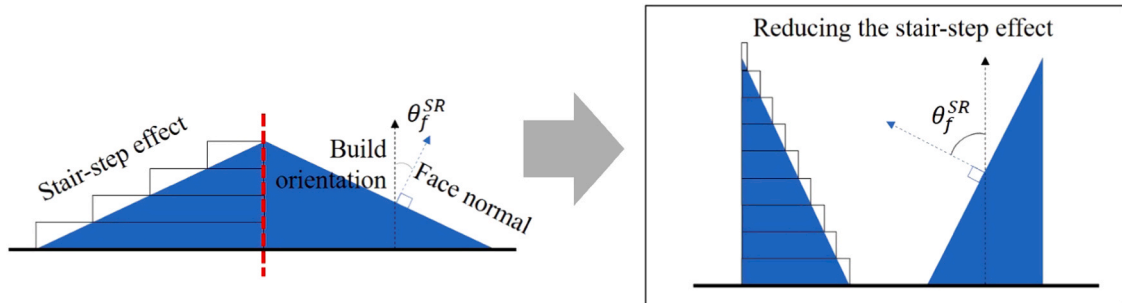


Fig. 2. Part decomposition to reduce the stair-step effect.

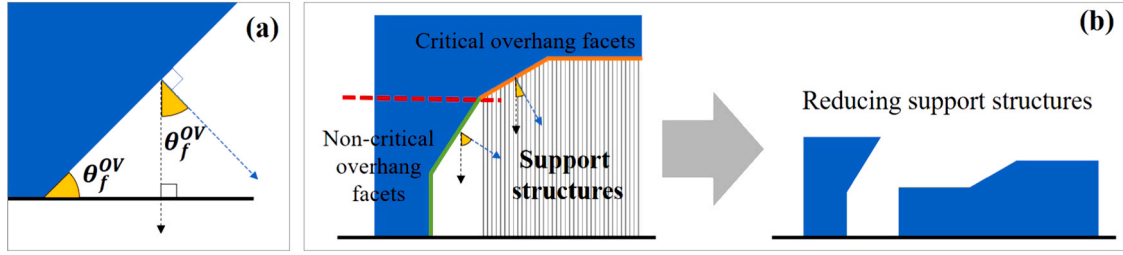


Fig. 3. (a) An angle of overhang facet f ; and (b) part decomposition to reduce support structures.

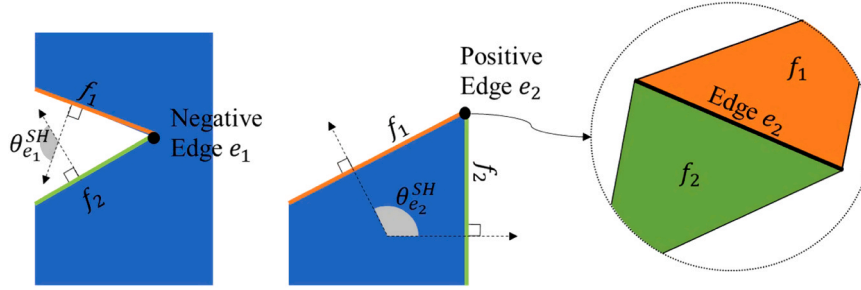


Fig. 4. Positive and negative edges.

where, θ_e^{SH} is the deviation angle between the normal vectors of the two contact facets of Edge e and its range is $0 \leq \theta_e^{SH} \leq \pi$; $L(e)$ is the length of Edge e ; and τ^{SH} is the threshold to determine whether a certain edge is sharp.

Minimizing G_3 can be leveraged as an objective function to remove sharp edges. As shown in Fig. 4, when $\theta_{e_1}^{SH}$ and $\theta_{e_2}^{SH}$ are beyond the threshold, τ^{SH} , Edges e_1 and e_2 become negative and positive sharp edges, respectively. When decomposing a part, it is likely that the number of positive edges increases while the number of negative edges decreases.

3.4. Gap: the area of facets causing non-allowable gaps

ISO/ASTM 52910 [15] refers to the minimum feature spacing as the minimum space that shall be specified between adjacent features. This consideration is addressed to ensure that the adjacent features do not fuse together during the build process. Herein, we name a space between adjacent features as a gap.

In most geometries, numerous gaps could be found. However, we only aim at identifying non-allowable gaps that are determined by a threshold that is a minimum distance between adjacent features. In this paper, we express a Gap, G_4 , as the sum of the areas of the facets causing non-allowable gaps for all decomposed components, as follows:

$$G_4(C) = \sum_{c \in C} \sum_{f \in c} E^{GA}(f) \quad (4)$$

$$E^{GA}(f) = \begin{cases} A(f) & \text{if } D(f, c) < \tau^{GA} \\ 0 & \text{otherwise} \end{cases}$$

where, τ^{GA} is the threshold to determine whether a gap is allowable; and $D(f, c)$ is the distance between Facet f and Component c .

A ray from a point on Facet f with the normal vector of Facet f is projected to Component c . If the ray hits a point on the surface of Component c , the distance is from the point on Facet f to the point of hit. The ray is originated from the center of a facet.

Minimizing G_4 can be leveraged as an objective function to remove non-allowable gaps. Fig. 5 shows how non-allowable gaps are removed through the PD process.

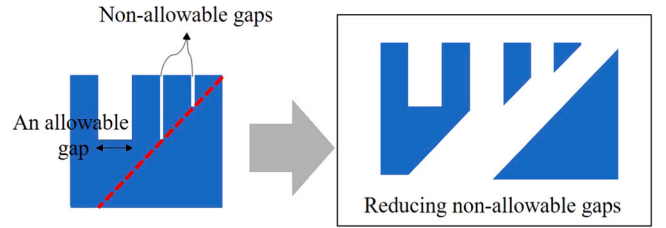


Fig. 5. Part decomposition to remove non-allowable gaps.

3.5. Concavity: the volume of concave spaces

ISO/ASTM 52910 [15] defines the maximum aspect ratio as the relationship between feature width and height or length. The maximum aspect ratio is taken into account to avoid features that are (i) too long or thin (e.g., thin walls [25]) and (ii) susceptible to failure during the build process. It is challenging to define a specific formula for the aspect ratio since there are a variety of feature-specific geometries. As such, we suggest a formulation that indirectly estimates the aspect ratio through the concept of Concavity. Concavity refers to a measure for concave features, such as sunken edges (see Fig. 6-(a)), hollows (see Fig. 6-(b)), and holes.

In the literature, there are primarily two types of formulations for Concavity: (1) length-based [10,30] and (2) volume-based [31]. The length-based formulation uses the longest projection distance that is from the farthest polygon to the convex hull. This could be useful to identify the best cutting region. However, this approach cannot address some types of concave features (e.g., hollows). Moreover, it could cause high computation complexity when a part has a significant number of facets in a mesh. On the other hand, the volume-based formulation uses the volume difference between an original model and its convex hull. This approach computes the remaining volume after subtracting the original model volume from its convex hull. Compared to the length-based formulation, the volume-based formulation is computationally efficient. As such, we adopt the volume-based approach to formulate Concavity. We refer to the remaining spaces after subtracting the original model from its convex hull as the *concave spaces*.

The aspect ratio of features is affected by the extent of protrusion that

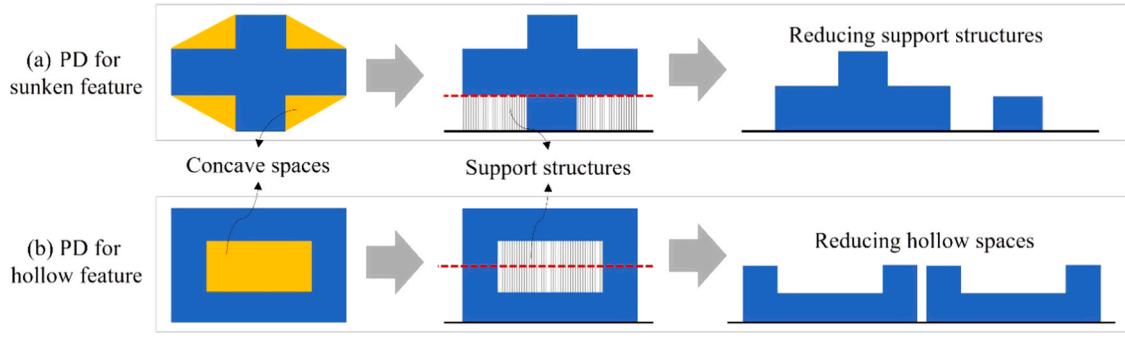


Fig. 6. Part decomposition to remove (a) sunken and (b) hollow features.

causes concave spaces. As such, it can be estimated by Concavity. Concavity, G_5 , is defined as the sum of the volumes of concave spaces for all decomposed components, which is expressed as follows:

$$G_5(C) = \sum_{c \in C} (V(H(c)) - V(c)) \quad (5)$$

where, $V(o)$ is the volume of Object o ; and $H(c)$ is the convex hull of Component c . Object o can be a convex hull or a component.

Similar to the aspect ratio, Concavity can be used to roughly estimate the overhang facets and feature spacing. This is because it is likely that concave spaces cause potential support structures and gaps. While it is challenging to more accurately estimate the potential issues (e.g., thin walls, support structures, and gaps), a main benefit of Concavity is that build orientations and user-defined thresholds are not required. To clarify, Concavity only requires geometry information. On the contrary, even though potential support structures can be measured by Eq. (2) of G_2 , the formulation requires information about the build orientation and geometry.

Minimizing G_5 can be leveraged as an objective function to remove concave spaces. Figure 6 (a) presents concave spaces that cause support structures and (b) demonstrates how hollow spaces are reduced by PD. As parts are more decomposed, G_5 decreases since decomposed parts move closer to convex hulls. Even when G_5 approaches zero, support structures can still be generated for some geometries (e.g., a sphere).

3.6. Feasibility: the fit feasibility of decomposed parts

As discussed earlier, fitting parts in the limited build envelope of an AM machine motivates implementing PD processes. In the literature, this is commonly referred to as *printability* [29,49] or *producibility* [6]. ISO/ASTM 52910 [15] addresses this consideration as the maximum part size meaning the largest part dimensions along the primary three axes as defined by ISO/ASTM 52921 [17] that can be produced in a given AM machine. In this paper, we refer to this design consideration as *Feasibility*, G_6 , of decomposed parts to ensure that the size of parts is smaller than the limited build envelope. A part that is smaller than the build envelope is called *feasible*. Otherwise, it is considered *infeasible*. Moreover, we accept the concept of ISO/ASTM 52910 [15] about the maximum part size.

Feasibility is affected by the build orientation of decomposed parts. Meanwhile, the build orientation might not be finally determined in the re-design stage for PD. In this case, a formulation for Feasibility could constrain the diagonal dimension of a part [35]. This is a way that a decomposed part could be fitted in the limited build space even if it is tilted. This can be expressed as follows:

$$G_6(C) = \begin{cases} 0 & \text{if } \sqrt{W^2 + L^2 + H^2} < \tau^{FE} \\ M & \text{otherwise} \end{cases} \quad (6)$$

where, τ^{FE} is the threshold to determine whether a decomposed part is feasible; W, L, H are the width, length, and height of the bounding box of

a component; and M is a significantly large value to avoid an undesired solution in a minimization problem. τ^{FE} could be differently defined depending on the shape (a cuboid or a cylinder) of the build space or no-build zones in the build volume [12]. If parts are built in a cuboid build space (i.e., a build envelope) without no-build zones, τ^{FE} could be the minimum of x-, y-, and z-dimensions of the build envelope [35].

Eq. (6) is formulated as an objective function for a minimization problem. Not only that, G_6 could also play a role as a constraint in the PD algorithm to ensure that decomposed parts are small enough. Namely, parts keep being decomposed in the PD algorithm until every component is deemed feasible [29]. In this case, $\sqrt{W^2 + L^2 + H^2} < \tau^{FE}$ could be used as a constraint.

3.7. Interface: the area of connection interface

ISO/ASTM 52910 [15] refers to assembly aspects as *recommended assembly fits*. This addresses the recommended clearance or interference between mating features. In this paper, we consider another assembly aspect, connection interface between sub-assemblies for assemblability. Generated by hyper-plane based decomposition, the connection interface means a set of facets where two components are in contact. In adhesive-based assembly, the connection interface is significant to ensure solid connection. *Interface*, G_7 , is the sum of the area of connection interface among decomposed components, which is expressed as follows:

$$G_7(C) = - \sum_{c \in C} \sum_{f \in I(c)} A(f)/2 \quad (7)$$

where, $I(c)$ is a set of facets included in the connection interface of Component c .

Since two decomposed parts share the same interface, G_7 is divided by two in Eq. (7). A significantly large interface between the two parts usually results in a solid connection, especially if the area of an interface is significant in the structural soundness of adhesive bonding. Therefore, maximizing G_7 could be implemented as an objective function. In our case, we put a negative sign in Eq. (7) to combine with other EIs as an objective function for minimization in Section 4.1. It should be noted that the concept of Interface is related to other design considerations for assemblability (e.g., ensuring a large enough interface for connector design [20]).

3.8. Quantity: the number of decomposed components

The number of parts is one of the factors to calculate assembly complexity [3]. ISO/ASTM 52910 [15] refers to part or product consolidation as a valuable design practice to minimize the number of parts. In this study, *Quantity*, G_8 , is the number of decomposed components and expressed as follows:

$$G_8(C) = |C| \quad (8)$$

where, the concatenation operation ($\|$) is used to provide the number of elements.

According to the *Design for Assembly (DFA)* principle, minimizing the number of parts can be implemented as an objective function to reduce assembly time and cost [3]. We include Quantity, the number of decomposed parts, as one of the key EIs to indirectly consider assembly time and cost. In the AM literature, the number of decomposed parts is often considered as one of the attributes in multi-attribute decision-making [29,33].

Even though the PD methodologies help to overcome the limitations of AM, they have a fundamental issue that the geometries of decomposed parts become simpler. This weakens one of the main advantages of AM enabling to produce complex parts. Therefore, it is significant to determine the proper number of decomposed parts [33].

4. Optimization of part decomposition

This section addresses the application of EIs to an optimization approach for PD. For the optimization approach, a *genetic algorithm (GA)* is adopted (Section 4.1). As a part of the GA, PD processes are included (Section 4.2).

4.1. A genetic algorithm

Fig. 7 represents the overall procedure of GA. With a set of input parameters, X , and an original model, PD processes in Part Decomposition are initiated (see Section 4.2). In Part Decomposition, the number and geometry of decomposed parts are decided from the input parameter set. The set of decomposed components, C , generated from the Part Decomposition method are assessed in Evaluation. Then a fitness value, Y , is calculated from Evaluation. Part Decomposition and Evaluation are repeated depending on the number of generations. To choose a solution (a chromosome in the GA) in a population set, the roulette-based selection is applied [41]. Through mutation and crossover, new chromosomes are created for each generation. The GA results in the optimal solution, X^* , that is the best set of input parameters to optimally decompose the original model.

In the Evaluation step, the set of decomposed components, C , are assessed based on a fitness function. The fitness function is a weighted sum as shown in Eq. (9). The proposed EIs are synthesized to form the fitness function that calculates the performance of a solution in the GA. In this paper, we minimize the fitness function that is an objective function. In Eq. (9), w_i and Z_i are a weight and a normalization function for i -th EI, respectively. In $Z_i(C)$, $G_i(C)$ falls between 0 and 1 with a normalization operation dividing by a standard value, ϵ_i . If a normalized outcome is too large, $G_i(C) > \epsilon_i$, or ϵ_i is 0, it is limited to 1. The standard values are the outcomes for EIs once V-HACD proceeds with the default input parameters.¹

$$\begin{aligned} \text{Min} : Y &= \sum_{i \in I} w_i Z_i(C) \\ Z_i(C) &= \begin{cases} 1 & \text{if } G_i(C) > \epsilon_i \text{ or } \epsilon_i = 0 \\ \frac{G_i(C)}{\epsilon_i} & \text{otherwise} \end{cases} \end{aligned} \quad (9)$$

The chromosome structure used for the GA is comprised of four genes: *Alpha*, *Beta*, *Gamma*, and *Maximum Concavity*. These are input parameters of the V-HACD algorithm [31]: *Alpha* controls the bias toward clipping along symmetry planes, *Beta* controls the bias toward clipping along revolution axes, *Gamma* is the maximum allowed concavity during the merge stage, and *Maximum Concavity* is a threshold that input primitives are recursively split until the concavity of each

subpart is lower than the threshold. According to the instructions of the V-HACD algorithm, the range of the four parameters is from 0 to 1. Additionally, the default values are 0.05, 0.05, 0.00125, and 0.0025, respectively.

4.2. Part decomposition processes

Section 4.2 describes the operations inside of the Part Decomposition box in Fig. 7. To decompose an original model into components, we apply two types of PD approaches: *concave feature-based PD (CPD)* and *convex feature-based PD (VPD)*. The V-HACD algorithm [31] is employed as a CPD approach. As explained in Section 2.2, V-HACD has the two critical issues: (1) the shape of decomposed parts becomes a convex hull; and (2) the algorithm cannot further decompose convex (or near-convex) geometries. To cope with these issues, we include follow-up processes after CPD (see Fig. 8). As shown in Fig. 9, an original model (StanfordBunny) is decomposed into convex hulls after CPD. To keep the original shape, we conduct a Boolean intersection between the original model and convex hulls. The Boolean intersection enables to conserve the original geometries as well as to keep the decomposed sections. Fig. 9(c) represents the outcome of the Boolean intersection and we name the output parts as pre-decomposed parts. Then, some pre-decomposed parts that are not feasible go through VPD. To determine Feasibility, Eq. (6) is used as a constraint. In the StanfordBunny example, even though the main body part (blue) is too large (not feasible), it cannot be decomposed in CPD since it is near-convex. Therefore, the part enters VPD and becomes two pieces as shown in Fig. 9(d). In the VPD step, an object is decomposed into two pieces by a cutting plane at the center of mass [36].

5. Case study

This section is comprised of three sub-sections. Section 5.1 shows how EIs are changed for the PD optimization. Section 5.2 represents how much build time and material consumption are reduced based on our PD approach. Section 5.3 provides a comparison of our approach and previous PD studies.

For these case studies, we implemented the PD and optimization algorithms in Python 3.7.4 and embedded them as add-in scripts in Blender 2.8.² Table 2 presents user-defined parameters including PD thresholds and GA parameters. Input 3D models are from the dataset of Chen et al. [8],³ NIST,⁴ and Thingiverse.⁵

5.1. Evaluation indicators for optimization

This sub-section addresses the PD optimization used at the re-design stage and a fitness value combining EIs is considered as a comparison indicator. To calculate certain EIs (i.e., Roughness, Overhang, and Feasibility), we need to determine build orientations. When deciding the build orientation of decomposed parts, we select the facet of the convex hull of the parts with the largest area as the base plane. However, when determining the build orientation of an original model for comparison, the initial orientation is used without re-orientation. Fig. 10 visually represents PD steps for a large model (Manhattan). The model size is $326 \times 542 \times 172$ mm which is larger than the allowed limit, 250 mm, which means that the original model is not feasible. Table 3 quantitatively details the change of EIs for each PD step. In Fig. 10(b), some pre-decomposed parts (especially the bottom plate) are too large to build in the workspace. Since it is not feasible, it is represented by M (see Eq. (6)) in Table 3. Infeasible parts are further decomposed into smaller pieces in

¹ <https://github.com/kmamou/v-hacd>

² <https://www.blender.org/>

³ <http://irc.cs.sdu.edu.cn/DAP/Dapper.html>

⁴ <https://pages.nist.gov/CAD-PMI-Testing/models.html>

⁵ <https://www.thingiverse.com>

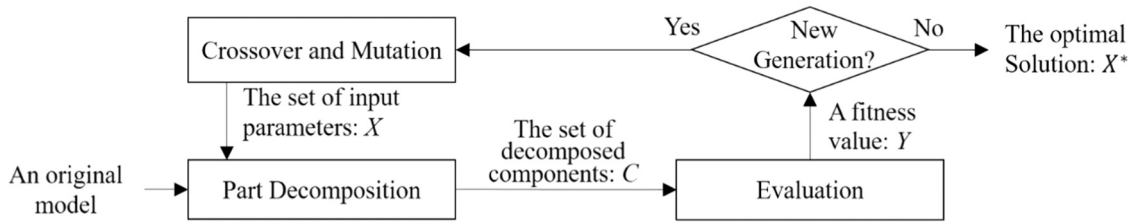


Fig. 7. A genetic algorithm for part decomposition.

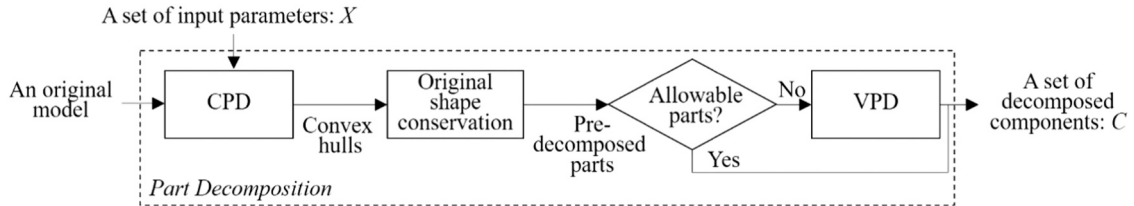


Fig. 8. Part decomposition processes including concave feature-based PD (CPD) and convex feature-based PD (VPD).

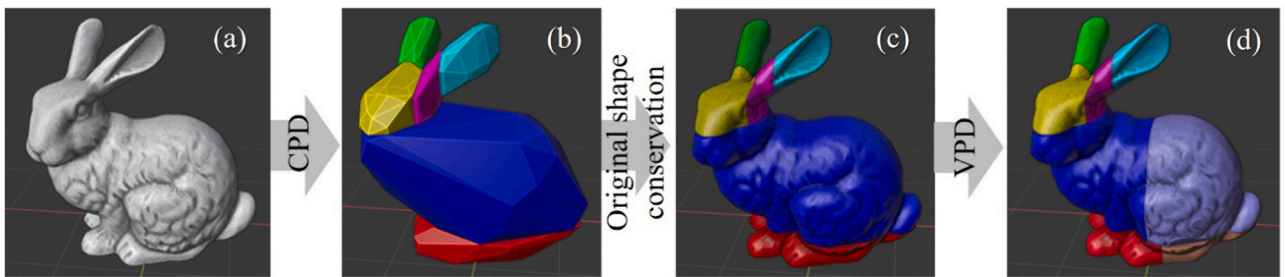


Fig. 9. Part decomposition processes: (a) an original model (StanfordBunny); (b) convex hulls; (c) pre-decomposed parts; and (d) decomposed parts.

Table 2
User-defined parameters.

	Parameters	Unit	Value
PD thresholds	τ^{OV} for critical overhang facets	Degree	60
	τ^{SH} for sharp edges	Degree	170
	τ^{GA} for non-allowable gaps	mm	2
	τ^{FE} for the feasibility of a part	mm	250
GA parameters	Number of populations	N/A	10
	Number of generations	N/A	300
	Probability of mutation	N/A	0.3

the next PD step (VPD). Then, every part becomes feasible as shown in Fig. 10-(c). These processes are repeated by changing a set of input parameters, X , for the PD algorithm. Through the GA, the set of input parameters is optimized to find the optimally decomposed parts. Fig. 10-(c) is an outcome of the PD algorithm with the default input parameters

of V-HACD. Since a relatively large number of decomposed parts are generated, it is adjusted through the GA-based optimization. In Table 3, this is witnessed by decreasing Quantity (41–33) at the expense of other EIs such as increasing Roughness (23,994–48,579). By reducing the number of decomposed parts in the middle of the PD optimization, the assembly time and cost could be saved. The outcomes of the optimization depend on the user-defined weighting of the EIs. In the optimization for the Manhattan model, the weights are set as $[0, 0, 0, 1, 1, 1, 0, 1]$ meaning that only the Gap, Concavity, Feasibility, and Quantity EIs are considered and are weighted equally. Fig. 11 shows another example that the GA-based optimization prevents over decomposition by constraining the number of decomposed parts with an EI, Quantity. In the middle of the optimization, a candidate with fewer decomposed parts is likely to have a better fitness value. Fig. 11-(b) is a PD result with the default input parameters, which shows that some parts (e.g., the circular bottom plate) are unnecessarily decomposed. After the GA-based

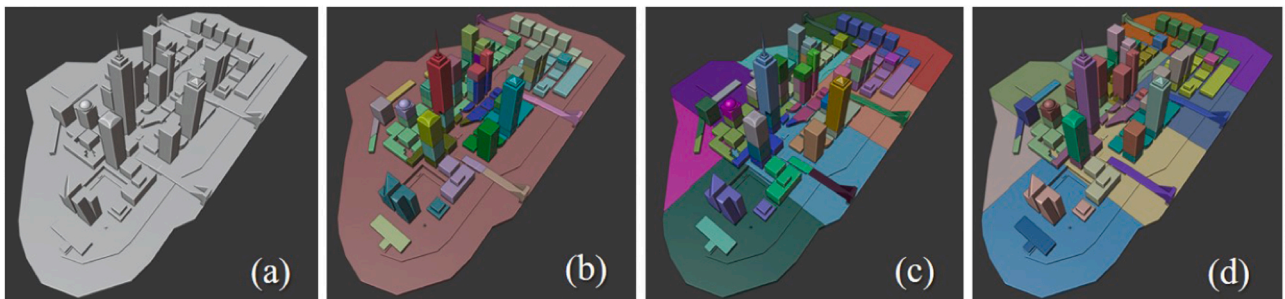


Fig. 10. Part decomposition steps for a large model (Manhattan): (a) an original model; (b) pre-decomposed parts; (c) decomposed parts; and (d) PD optimization.

Table 3
Evaluation Indicators for part decomposition steps in Fig. 10.

	(a) An original model	(b) Pre-decomposed parts	(c) Decomposed parts	(d) PD Optimization
Roughness	3114	23,994	23,994	48,579
Overhang	732	9609	9609	22,387
Sharpness	736	839	839	843
Gap	96	99	99	99
Concavity	8,331,363	1,561,728	1,113,136	1,147,910
Feasibility	M	M	0	0
Interface	0	2965	9696	8021
Quantity	1	33	41	33

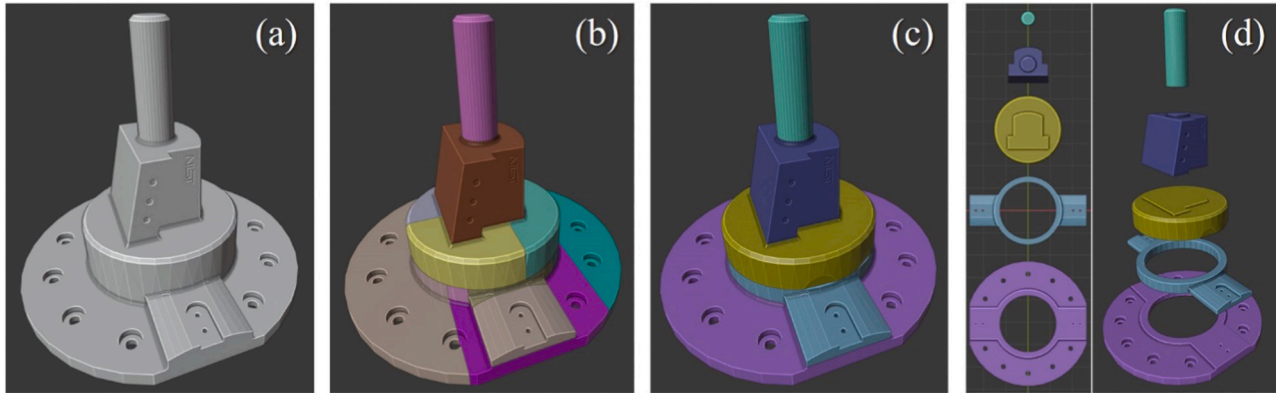


Fig. 11. Comparison between default PD and optimized PD: (a) original model (NistPart1); (b) PD with default parameters; (c) PD with optimized input parameters; and (d) different views.

optimization, only five decomposed parts are generated as shown in Fig. 11-(c) and (d).

5.2. Comparison with original models

The two build cases of an original model and decomposed parts are compared under different AM processes: (1) *material extrusion (ME)* and (2) *vat photopolymerization (VP)*. For each AM process, commercial slicing applications are employed: *Ultimaker Cura 4.5*⁶ for ME and *PrusaSlicer 2.2.0*⁷ for VP. To estimate build time and material consumption, virtual AM machines are embedded in slicing applications. For our demonstration, *Ultimaker Original* and *Original Prusa SL1* are used as ME and VP AM machines, respectively. The main process parameters of the virtual AM machines for the case studies are shown in Table 4. For process parameters, the default values of the virtual machines are mostly used.

Table 4
The main process parameters of the virtual AM machines.

	Ultimaker Original	Original Prusa SL1
Size of build envelope: width × length × height (mm)	200 × 200 × 300	200 × 200 × 300
Layer thickness (mm)	0.2	0.05
Infill rate (%)	20	50
Support pattern	Zig-Zag	Zig-Zag
Support overhang angle (Degrees)	50	45

In the case studies, while Section 5.1 deals with how to decompose at the re-design stage, Sections 5.2 and 5.3 address how to build at the process planning stage. Build orientations can be determined at both re-design and process planning stages. As the build orientation is one of the process parameters, it is usually determined at the process planning stage. However, when considering build orientations at the re-design stage, we can save build time and cost for AM by efficiently re-designing a model. However, it is challenging to find the optimal build orientation at the re-design stage by considering all process parameters. Therefore, the build orientation decided at the early stage can be tuned at the process planning stage. In Section 5.1, we determined the build orientation for certain EIs (i.e., Roughness, Overhang, and Feasi-

bility) to evaluate decomposed parts by selecting the largest facet of a convex hull. On the other hand, in Sections 5.2 and 5.3, when decomposed parts are built, we use the slicing applications' functionality that is automatically determining the orientation and location of the decomposed parts on the build platform. In Sections 5.2 and 5.3, complicated issues (e.g., considering multiple builds [35] and 3D packing that is stacking parts over other parts [11]) in process planning for AM [34] are simplified by building all decomposed parts in the same build and applying 2D placement [50].

As shown in Table 5, compared to an original model, decomposed parts generated from the optimization usually have less build time and material consumption. This is mainly because the amount of support structure is reduced by decomposing an original model and re-orienting decomposed parts. Additionally, build height reduction is usually critical for VP in decreasing the build time. Fig. 12 presents the PD outcomes of the optimization for various models used in the comparison. Our approach applied to the *Chair* model (see Fig. 13) demonstrates that decomposed parts dramatically reduce the amount of support structure and the build height. When decomposed parts are built for both ME and VP, almost no support structure is generated. As a result, the build time decreases by 31% for ME and 90% for VP and the material consumption is reduced by 38% for ME and 32% for VP. Note that PD does not always bring positive results (see NistPart1 in Table 5). This is mainly because decomposed parts could cause more support structures depending on their build orientations.

5.3. Comparison with the previous studies

Compared with previous studies, we provide an advanced PD approach. Fig. 14 shows the outcomes of the decomposing the *Inukshuk* model using different PD approaches. *Dapper* [8] is challenging to elaborately decompose concave features that usually cause manufacturability issues such as support structure. As such, when the model is

⁶ <https://ultimaker.com/software/ultimaker-cura>

⁷ <https://www.prusa3d.com/prusaslicer>

Table 5

Comparison of (a) original and (b) decomposed models based on build time (BT) and material consumption (MC) for material extrusion (ME) and vat photopolymerization (VP).

Model Name	ME (BT: min; MC: g)						VP (BT: min; MC: ml)					
	(a)		(b)		(b)/ (a)		(a)		(b)		(b)/ (a)	
	BT	MC	BT	MC	BT	MC	BT	MC	BT	MC	BT	MC
Boards	119	15	93	11	0.78	0.73	28	13	12	10	0.43	0.77
Chair	203	29	141	18	0.69	0.62	67	38	7	26	0.10	0.68
GoldenGateBridge	162	15	96	10	0.59	0.67	36	19	13	13	0.36	0.68
NistPart1	83	10	88	11	1.06	1.10	41	12	13	13	0.32	1.08
NistPart2	83	11	82	10	0.99	0.91	13	20	9	14	0.69	0.70
Plane	151	21	140	17	0.93	0.81	20	36	8	26	0.40	0.72
Statue	178	23	169	22	0.95	0.96	99	46	21	48	0.21	1.04
Table	180	27	104	12	0.58	0.44	36	27	5	11	0.14	0.41
TallChair	354	61	191	21	0.54	0.34	130	39	9	36	0.07	0.92

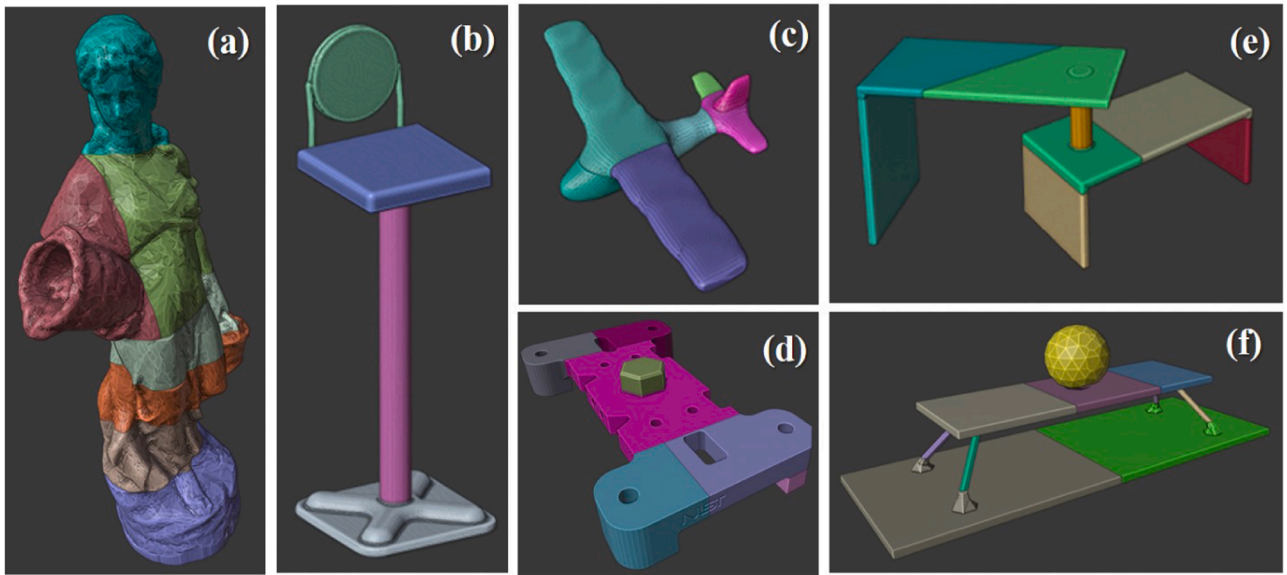


Fig. 12. Other cases: (a) Statue; (b) TallChair; (c) Plane; (d) NistPart2; (e) Table; and (f) Boards.

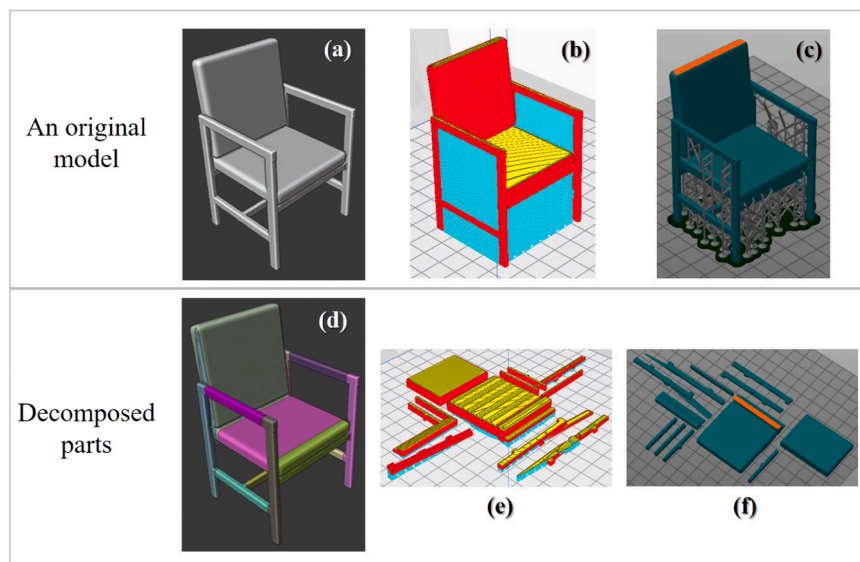


Fig. 13. Build processes for (a) an original model (Chair) and (d) decomposed parts: (b, e) material extrusion; and (c, f) vat photopolymerization.

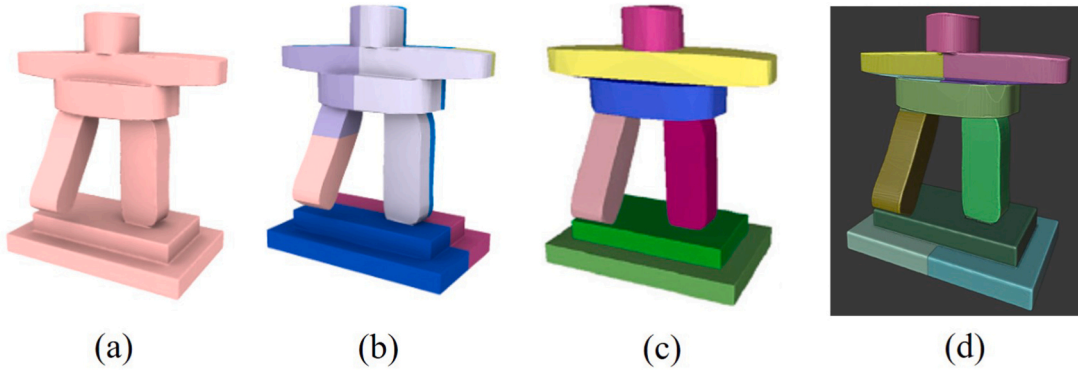


Fig. 14. Inukshuk: (a) original; (b) Dapper [8]; (c) Near-convex decomposition [10]; and (d) our approach.

decomposed, concave edges are not focused to cut (see, Fig. 14-(b)). *Near-convex decomposition (NCD)* [10] provides a more advanced decomposition by addressing concavity for PD (see, Fig. 14-(c)). However, this method cannot decompose "near" convex parts into smaller pieces. This could be problematic if convex or near-convex parts are larger than the build envelope of an AM machine, which is infeasible. On the other hand, we provide the PD approach for convex parts that is VPD in Section 4.2. Feasibility is considered as a constraint in the decomposition step for convex parts. In this example, we set the Inukshuk model is larger than the build envelope and decomposed it. As shown in Fig. 14-(d), two large convex components (the baseplate and the arms) that are not feasible are decomposed into two pieces, respectively.

In comparison to NCD [10] that adopted the same CPD algorithm (V-HACD), our approach is more effective in reducing material consumption. This is mainly because our approach obtains original geometries from convex hulls while NCD considers convex hulls as final

decomposed parts. In the comparison experiment, the same input parameters of V-HACD that are used in the BOBYQA-based optimization for NCD. Fig. 15 visually compares how original models (Eiffel-Tower, Lattice, and Honeycomb) are decomposed using NCD and our approach. The amount of consumed material is shown in Fig. 16. For the Honeycomb model, the material consumption is reduced by 26% for material extrusion (ME) and 62% for vat photopolymerization (VP). Our approach can reduce the amount of material by reducing the unnecessary volume generated by convex hulls. The material consumption gap between the two approaches tends to increase when the original model includes lattice structures.

6. Conclusion and future work

This study is an attempt to incorporate the standard design guidelines for AM into PD. A variety of design considerations can be addressed

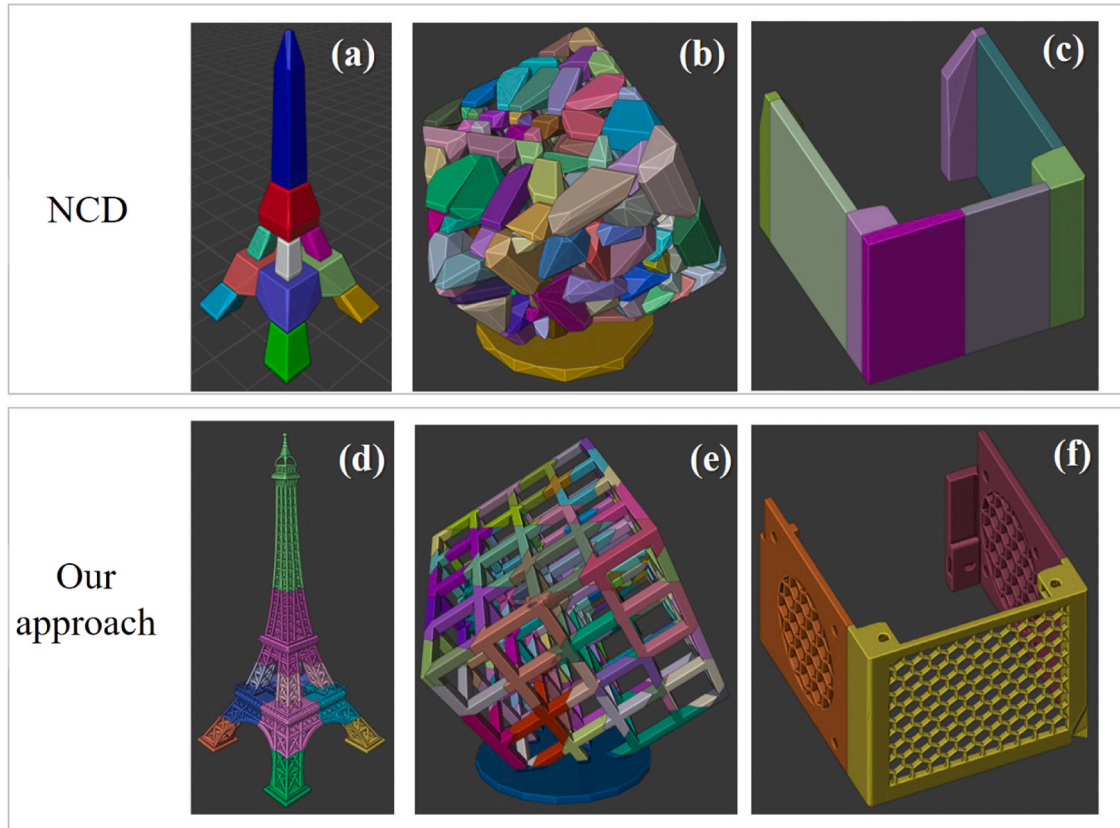


Fig. 15. Comparison of near-convex decomposition (NCD) [10] and our approach: (a, d) EiffelTower; (b, e) Lattice; and (c, f) Honeycomb.

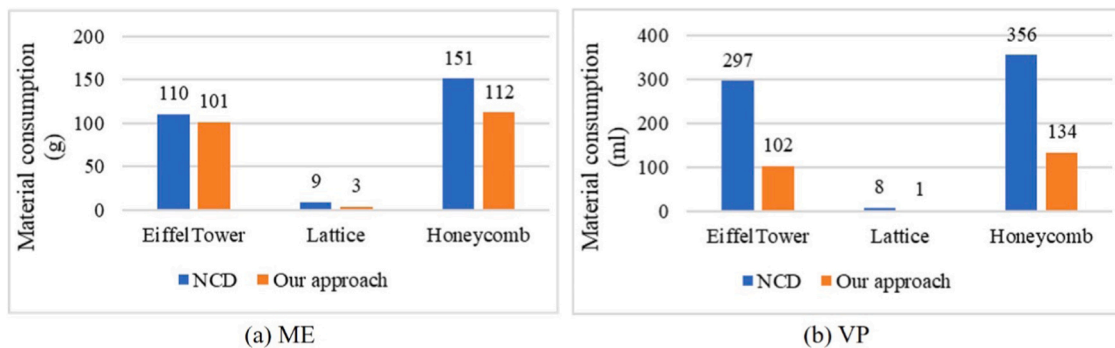


Fig. 16. Material consumption comparison of near-convex decomposition (NCD) [10] and our approach: (a) Material Extrusion (ME); and (b) Vat Photopolymerization (VP).

to evaluate that an original model is properly decomposed. However, PD studies have often proposed their ad-hoc design considerations to achieve certain objectives. In this paper, we develop a comprehensive approach that addresses common and essential design considerations using the standard AM design guidelines found in ISO/ASTM 52910. We classify the design evaluation indicators (EIs) found in the standard into manufacturability and assemblability concerns and develop mathematical expressions for eight EIs. The mathematical expressions for these EIs enable them to be incorporated into the part decomposition optimization method. A generic algorithm is adopted for the optimization approach and PD processes including CPD and VPD are considered to decompose an original model. In the case studies, decomposed parts are compared with their original models in terms of build time and material consumption, and the performance of our optimization approach is compared with previous studies.

For future work, this paper leaves further issues spanning some research directions. First, although the design considerations could be mathematically defined in various ways, we aimed at providing rather simplified formulas with essential variables for each design consideration. Therefore, the evaluation indicators could be extended or re-defined with other perspectives. For example, the concept of sharpness can be re-defined as vertices rather than edges. Second, some guidelines to suggest proper evaluation indicators for PD could be provided depending on a variety of AM processes. This is mainly because the EIs for PD could be selectively applied depending on AM processes. For instance, an evaluation indicator for Overhang could be ignored when an AM process does not generate support structure. Third, although this paper is based on adhesive-based assembly, other assembly types could be considered. If fastener-based assembly is considered, incorporating the connectors (e.g., numbers, shape, or angles) will affect evaluation indicators for assemblability. Fourth, depending on a variety of conditions (e.g., material types, build orientations, and AM process types), typical PD thresholds for EI could be investigated by proceeding practical experiments with test specimens. Fifth, the GA-based optimization could be improved by changing conditions (e.g., tournament-based selection) and parameters (e.g., number of populations or generations) or replaced with other meta-heuristic methods (e.g., simulated annealing) to reduce computation time or to increase the performance. Lastly, in this paper, the objective function for the optimization is synthesized by the EIs, which is multi-attribute decision making. However, since an individual objective is assigned for each EI, the relations of EIs could be analyzed in terms of trade-offs among conflict objectives.

CRedit authorship contribution statement

Yosep Oh: Conceptualization, Methodology, Investigation, Formal analysis, Software, Validation, Data Curation, Writing - Original Draft, Writing - Review & Editing. **Hyunwoong Ko:** Conceptualization, Methodology, Writing - Original Draft. **Timothy Sprock:** Methodology,

Writing - Original Draft. **William Z. Bernstein:** Methodology, Writing - Original Draft. **Soonjo Kwon:** Conceptualization, Methodology, Formal analysis, Data Curation, Supervision, Project administration, Writing - Original Draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We thank Allison Barnard Feeney, Paul Witherell and Vijay Srinivasan at NIST, and Dongmin Shin at Hanyang University, and the anonymous peer-reviewers for their valuable comments and input.

Disclaimer

No endorsement of any commercial product by NIST is intended. Commercial materials are identified in this report to facilitate better understanding. Such identification does not imply endorsement by NIST nor does it imply the materials identified are necessarily the best for the purpose.

References

- [1] P. Alexander, S. Allen, D. Dutta, Part orientation and build cost determination in layered manufacturing, *Comput.-Aided Des.* 30 (1998) 343–356.
- [2] H. Bikas, A. Lianos, P. Stavropoulos, A design framework for additive manufacturing, *Int. J. Adv. Manuf. Technol.* 103 (2019) 3769–3783.
- [3] G. Boothroyd, Product design for manufacture and assembly, *Comput.-Aided Des.* 26 (1994) 505–520.
- [4] H.S. Byun, K.H. Lee, Determination of the optimal part orientation in layered manufacturing using a genetic algorithm, *Int. J. Prod. Res.* 43 (2005) 2709–2724.
- [5] F. Calignano, Design optimization of supports for overhanging structures in aluminum and titanium alloys by selective laser melting, *Mater. Des.* 64 (2014) 203–213.
- [6] C. Chan, S. Tan, Volume decomposition of cad models for rapid prototyping technology, *Rapid Prototyp. J.* (2005).
- [7] B. Chazelle, Convex partitions of polyhedra: a lower bound and worst-case optimal algorithm, *SIAM J. Comput.* 13 (1984) 488–507.
- [8] X. Chen, H. Zhang, J. Lin, R. Hu, L. Lu, Q.X. Huang, B. Benes, D. Cohen-Or, B. Chen, Dapper: decompose-and-pack for 3d printing, *ACM Trans. Graph.* 34 (2015), 213–1.
- [9] T.H. Cormen, C.E. Leiserson, R.L. Rivest, C. Stein, *Introduction to Algorithms*, MIT Press, 2009.
- [10] I. Demir, D.G. Aliaga, B. Benes, Near-convex decomposition and layering for efficient 3d printing, *Addit. Manuf.* 21 (2018) 383–394.
- [11] A. Gogate, S. Pande, Intelligent layout planning for rapid prototyping, *Int. J. Prod. Res.* 46 (2008) 5607–5631.
- [12] V. Griffiths, J.P. Scanlan, M.H. Eres, A. Martinez-Sykora, P. Chinchapatnam, Cost-driven build orientation and bin packing of parts in selective laser melting (slm), *Eur. J. Oper. Res.* 273 (2019) 334–352.
- [13] ISO4287, Geometrical Product Specifications (GPS) — Surface texture: Profile method — Terms, definitions and surface texture parameters, 1997.

- [14] ISO/ASTM52900, Standard Terminology for Additive Manufacturing — General Principles — Terminology, 2016.
- [15] ISO/ASTM52910, Additive manufacturing — Design — Requirements, guidelines and recommendations, 2018.
- [16] ISO/ASTM52911-2, Additive manufacturing — Design — Part 2: Laser-based powder bed fusion of polymers, 2019.
- [17] ISO/ASTM52921, Standard Terminology for Additive Manufacturing-Coordinate Systems and Test Methodologies, 2013.
- [18] J.P. Järvinen, V. Matilainen, X. Li, H. Piili, A. Salminen, I. Mäkelä, O. Nyrhilä, Characterization of effect of support structures in laser additive manufacturing of stainless steel, *Phys. Procedia* 56 (2014) 72–81.
- [19] J. Jiang, X. Xu, J. Stringer, Support structures for additive manufacturing: a review, *J. Manuf. Mater. Process.* 2 (2018) 64.
- [20] X. Jiang, X. Cheng, Q. Peng, L. Liang, N. Dai, M. Wei, C. Cheng, Models partition for 3d printing objects using skeleton, *Rapid Prototyp. J.* 23 (2017) 54–64.
- [21] E. Karasik, R. Fattal, M. Werman, Object partitioning for support-free 3d-printing, *Comput. Graph. Forum* (2019) 305–316.
- [22] H. Ko, S.K. Moon, J. Hwang, Design for additive manufacturing in customized products, *Int. J. Precis. Eng. Manuf.* 16 (2015) 2369–2375, <https://doi.org/10.1007/s12541-015-0305-9>.
- [23] H. Ko, S.K. Moon, K.N. Otto, Design knowledge representation to support personalised additive manufacturing, *Virtual Phys. Prototyp.* 10 (2015) 217–226, <https://doi.org/10.1080/17452759.2015.1107942>.
- [24] H. Ko, P. Witherell, Y. Lu, S. Kim, D.W. Rosen, Machine learning and knowledge graph based design rule construction for additive manufacturing, *Addit. Manuf.* (2020).
- [25] J. Kranz, D. Herzog, C. Emmelmann, Design guidelines for laser additive manufacturing of lightweight structures in tial6v4, *J. Laser Appl.* 27 (2015), S14001.
- [26] S. Kwon, D. Mun, B.C. Kim, S. Han, H.W. Suh, B-rep model simplification using selective and iterative volume decomposition to obtain finer multi-resolution models, *Comput.-Aided Des.* 112 (2019) 23–34.
- [27] J.M. Lien, N.M. Amato, 2007. Approximate convex decomposition of polyhedra, in: *Proceedings of the 2007 ACM symposium on Solid and physical modeling*, 121–131.
- [28] J.M. Lien, N.M. Amato, Approximate convex decomposition of polyhedra and its applications, *Comput. Aided Geom. Des.* 25 (2008) 503–522.
- [29] L. Luo, I. Baran, S. Rusinkiewicz, W. Matusik, Chopper: partitioning models into 3d-printable parts, *ACM Trans. Graph.* 31 (2012) 1–9.
- [30] K. Mamou, F. Ghorbel, A simple and efficient approach for 3d mesh approximate convex decomposition, in: *2009 16th IEEE international conference on image processing (ICIP)*, IEEE, 2009, 3501–3504.
- [31] K. Mamou, E. Lengyel, A. Peters, Volumetric hierarchical approximate convex decomposition, *Game Engine Gems 3*, 2016, pp. 141–158.
- [32] M. Meyer, M. Desbrun, P. Schröder, A.H. Barr, *Discrete differential-geometry operators for triangulated 2-manifolds. Visualization and Mathematics III*, Springer, 2003, pp. 35–57.
- [33] Y. Oh, S. Behdad, C. Zhou, Part separation methods for assembly based design in additive manufacturing, in: *ASME 2017 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, American Society of Mechanical Engineers Digital Collection, 2017, V004T05A014.
- [34] Y. Oh, P. Witherell, Y. Lu, T. Sprock, Nesting and scheduling problems for additive manufacturing: a taxonomy and review, *Addit. Manuf.* (2020), 101492.
- [35] Y. Oh, C. Zhou, S. Behdad, Part decomposition and 2d batch placement in single-machine additive manufacturing systems, *J. Manuf. Syst.* 48 (2018) 131–139.
- [36] Y. Oh, C. Zhou, S. Behdad, Part decomposition and assembly-based (re) design for additive manufacturing: a review, *Addit. Manuf.* 22 (2018) 230–242.
- [37] M.J. Powell, *The Broyden Algorithm for Bound Constrained Optimization Without Derivatives*, University of Cambridge, Cambridge, 2009, pp. 26–46. Cambridge NA Report NA2009/06.
- [38] B.S. Rupal, N. Anwer, M. Secanell, A.J. Qureshi, Geometric tolerance and manufacturing assemblability estimation of metal additive manufacturing (am) processes, *Mater. Des.* (2020), 108842.
- [39] H. Sakurai, Volume decomposition and feature recognition: part 1-polyhedral objects, *Comput.-Aided Des.* 27 (1995) 833–843.
- [40] J. Schmelzle, E.V. Kline, C.J. Dickman, E.W. Reutzel, G. Jones, T.W. Simpson, (re) designing for part consolidation: understanding the challenges of metal additive manufacturing, *J. Mech. Des.* 137 (2015).
- [41] A. Shukla, H.M. Pandey, D. Mehrotra, Comparative review of selection techniques in genetic algorithm, in: *2015 International Conference on Futuristic Trends on Computational Analysis and Knowledge Management (ABLAZE)*, IEEE, 2015, 515–519.
- [42] M. Szilvási-Nagy, About curvatures on triangle meshes, *KoG* 10 (2006) 13–18.
- [43] M.K. Thompson, G. Moroni, T. Vaneker, G. Fadel, R.I. Campbell, I. Gibson, A. Bernard, J. Schulz, P. Graf, B. Ahuja, et al., Design for additive manufacturing: Trends, opportunities, considerations, and constraints, *CIRP Ann.* 65 (2016) 737–760.
- [44] T. Vaneker, A. Bernard, G. Moroni, I. Gibson, Y. Zhang, Design for additive manufacturing: framework and methodology, *CIRP Ann.* (2020).
- [45] W.M. Wang, C. Zanni, L. Kobbelt, Improved surface quality in 3d printing by optimizing the printing direction, *Computer Graphics Forum*, Wiley Online Library, 2016, pp. 59–70.
- [46] G. Williams, N.A. Meisel, T.W. Simpson, C. McComb, Design repository effectiveness for 3d convolutional neural networks: application to additive manufacturing, *J. Mech. Des.* 141 (2019).
- [47] C. Wu, C. Dai, G. Fang, Y.J. Liu, C.C. Wang, General support-effective decomposition for multi-directional 3-d printing, *IEEE Trans. Autom. Sci. Eng.* (2019).
- [48] S. Yang, Y. Tang, Y.F. Zhao, A new part consolidation method to embrace the design freedom of additive manufacturing, *J. Manuf. Process.* 20 (2015) 444–449.
- [49] M. Yao, Z. Chen, L. Luo, R. Wang, H. Wang, Level-set-based partitioning and packing optimization of a printable model, *ACM Trans. Graph.* 34 (2015) 1–11.
- [50] Y. Zhang, R.K. Gupta, A. Bernard, Two-dimensional placement optimization for multi-parts production in additive manufacturing, *Robot. Comput.-Integr. Manuf.* 38 (2016) 102–117.